

# EEB3 SOLID-STATE CLOCK

## Hardware Wiring & Connection Guide (Parallel Power & Opto-Isolation)

### 1. POWER WIRING (5V / 3A Regulated Parallel Power Setup)

**Wiring Goal:** Power the Arduino Mega and the 4 relay modules from the same 5V adapter. To isolate the Arduino from coil noise, **remove the yellow jumpers** on all relay boards.

| Connection Line         | From                        | To Pin                                                                 | Connection Notes / Details                                                  |
|-------------------------|-----------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Main Adapter Power (+)  | External 5V (+) Power Rail  | Mega 5V (Header Pin) & all 4 relay boards JD-VCC pins                  | <b>CRITICAL:</b> Yellow JD-VCC jumper <b>MUST</b> be removed on all boards. |
| Main Adapter Ground (-) | External 5V (-) Ground Rail | Mega GND (Header Pin) & all 4 relay boards GND pins (on 3-pin headers) | Provides a clean common return path for both logic and coils.               |
| Optocoupler Power       | Arduino Mega 5V             | All 4 relay boards VCC pins (on 10-pin headers)                        | Powers the opto-isolation diodes from the Mega's regulated 5V line.         |
| Signal Control Lines    | Arduino Mega Pins 22-53     | IN1 to IN8 on 10-pin headers                                           | Do NOT bridge Arduino GND to the 10-pin header GND. Leave it disconnected.  |

### 2. DS3231 RTC (Real-Time Clock) CONNECTIONS

**Note:** Wire the RTC DS3231 module directly to the Arduino Mega using the hardware I2C pins. Pins 20 and 21 are the *only* hardware I2C pins on the Mega.

| RTC Pin | Arduino Mega Pin       | Function     | Connection Details / Notes                                              |
|---------|------------------------|--------------|-------------------------------------------------------------------------|
| VCC     | Mega 5V (POWER header) | Logic Supply | Draws only ~2 mA. Completely safe to power from Arduino Mega's 5V line. |
| GND     | Mega GND               | Ground       | Common ground reference.                                                |
| SDA     | Mega Pin 20            | I2C Data     | Hardware SDA on Arduino Mega (do not use Uno A4).                       |
| SCL     | Mega Pin 21            | I2C Clock    | Hardware SCL on Arduino Mega (do not use Uno A5).                       |

### 3. RELAY BOARD TO BULB MAIN SIDE (220V AC WIRING)

| Relay Terminal       | Wire Connects To      | Relay State OFF                    | Relay State ON                     |
|----------------------|-----------------------|------------------------------------|------------------------------------|
| COM (Common)         | Mains 220V Live Wire  | Mains connected to common terminal | Mains connected to common terminal |
| NO (Normally Open)   | Light Bulb Terminal A | Open Circuit (Bulb is OFF)         | Closed Circuit (Bulb is ON)        |
| NC (Normally Closed) | Leave Unconnected     | Closed Circuit                     | Open Circuit                       |

**WARNING:** 220V wiring carries lethal voltages. All mains connections must be fused, insulated, and enclosed in a protective chassis. Ensure mains power is disconnected before wiring.

#### 4. SIGNAL CONTROL WIRE PINMAPS (Arduino Mega Pins 22 to 53)

The signal cables connect the Arduino Mega's double-row header to the 10-pin header on each relay board.

##### Module 1 (Hours Tens) & Module 2 (Hours Units)

| Relay IN | Module 1 Pin | Segment          | Frame | Relay IN | Module 2 Pin | Segment          | Frame |
|----------|--------------|------------------|-------|----------|--------------|------------------|-------|
| IN1      | Pin 22       | f (Top-Left)     | 1.1   | IN1      | Pin 30       | f (Top-Left)     | 2.1   |
| IN2      | Pin 23       | a (Top)          | 1.2   | IN2      | Pin 31       | a (Top)          | 2.2   |
| IN3      | Pin 24       | b (Top-Right)    | 1.3   | IN3      | Pin 32       | b (Top-Right)    | 2.3   |
| IN4      | Pin 25       | g (Middle)       | 1.4   | IN4      | Pin 33       | g (Middle)       | 2.4   |
| IN5      | Pin 26       | e (Bottom-Left)  | 1.5   | IN5      | Pin 34       | e (Bottom-Left)  | 2.5   |
| IN6      | Pin 27       | c (Bottom-Right) | 1.6   | IN6      | Pin 35       | c (Bottom-Right) | 2.6   |
| IN7      | Pin 28       | d (Bottom)       | 1.7   | IN7      | Pin 36       | d (Bottom)       | 2.7   |
| IN8      | Pin 29       | Colon Dots       | 1.8   | IN8      | Pin 37       | Unused / Empty   | —     |

##### Module 3 (Minutes Tens) & Module 4 (Minutes Units)

| Relay IN | Module 3 Pin | Segment          | Frame | Relay IN | Module 4 Pin | Segment          | Frame |
|----------|--------------|------------------|-------|----------|--------------|------------------|-------|
| IN1      | Pin 38       | f (Top-Left)     | 3.1   | IN1      | Pin 53       | f (Top-Left)     | 4.1   |
| IN2      | Pin 39       | a (Top)          | 3.2   | IN2      | Pin 52       | a (Top)          | 4.2   |
| IN3      | Pin 40       | b (Top-Right)    | 3.3   | IN3      | Pin 48       | b (Top-Right)    | 4.3   |
| IN4      | Pin 41       | g (Middle)       | 3.4   | IN4      | Pin 50       | g (Middle)       | 4.4   |
| IN5      | Pin 42       | e (Bottom-Left)  | 3.5   | IN5      | Pin 49       | e (Bottom-Left)  | 4.5   |
| IN6      | Pin 43       | c (Bottom-Right) | 3.6   | IN6      | Pin 51       | c (Bottom-Right) | 4.6   |
| IN7      | Pin 44       | d (Bottom)       | 3.7   | IN7      | Pin 46       | d (Bottom)       | 4.7   |
| IN8      | Pin 45       | Unused / Empty   | —     | IN8      | Pin 47       | Unused / Empty   | —     |

**Module 4 wiring-swap fix:** unlike Modules 1-3, Module 4's connector pins are NOT in sequential IN1-IN8 order. IN3 is on Pin 48 and IN6 is on Pin 51 — the reverse of what a naive sequential wiring would suggest. This was a confirmed jumbled-connector fix; wire Module 4 exactly as tabulated above, not by pin-number order.

#### 5. SEGMENT REF MAP (SEG2RELAY[7] = { 1, 2, 5, 6, 4, 0, 3 })

The software controls digits by mapping standard segment indices (0-6 corresponding to a-g) to relay channels. The segment locations on the physical frame are shown below.

| Segment Index | Letter | Relay Input Channel | Standard Position on Clock Frame |
|---------------|--------|---------------------|----------------------------------|
| 0             | a      | IN2                 | Top horizontal segment           |
| 1             | b      | IN3                 | Top-right vertical segment       |
| 2             | c      | IN6                 | Bottom-right vertical segment    |
| 3             | d      | IN7                 | Bottom horizontal segment        |
| 4             | e      | IN5                 | Bottom-left vertical segment     |
| 5             | f      | IN1                 | Top-left vertical segment        |
| 6             | g      | IN4                 | Middle horizontal segment        |

## 6. WIRING VERIFICATION — DIAGNOSTIC MODE

Once all signal and mains connections above are made, use the firmware's built-in diagnostic protocol to confirm every relay fires the correct bulb before buttoning up the enclosure — no need to wait for a particular time of day to test a segment.

| Serial Command | Effect                                                                                                                                    |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| D1             | Enter diagnostic mode — clock display pauses, all relays forced OFF.                                                                      |
| D0             | Exit diagnostic mode — normal HH:MM clock display resumes.                                                                                |
| R<m><i><s>     | Fire one relay directly: m = module 0-3, i = relay index 0-7 (IN1=0 ... IN8=7), s = 0/1. Example: R071 turns Module 0 IN8 (the colon) ON. |

**clock\_diagnostics.html** (in the EEB3\_Clock\_Mega\_Relay\_PowerSaving folder) wraps this protocol in a browser UI: open it in Chrome or Edge (Web Serial required), click Connect and pick the Arduino's serial port, and it automatically sends D1 and exposes a Test button for every relay on every module — each press sends the matching R<m><i><s> command so you can trace bulb-to-relay wiring one channel at a time. Disconnecting sends D0 to resume the clock.